

Agent-Client Matching Model

github.com/zheyuanlai/singlife-datathon

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Outline

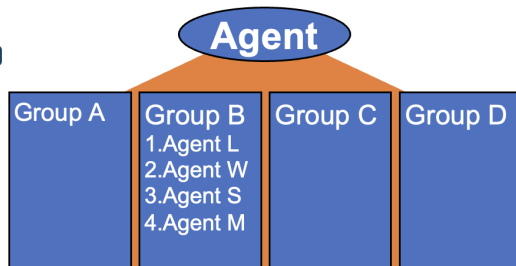
- 1 Introduction
- 2 Dataset Overview
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Introduction

- **Objective:** Develop a data-driven model to recommend the best financial agents to customers.
- **Goal:** Enhance policy conversion rates, customer satisfaction, and revenue.
- **Pipeline:** cluster agents, assign groups, and rank agents.



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Dataset Overview

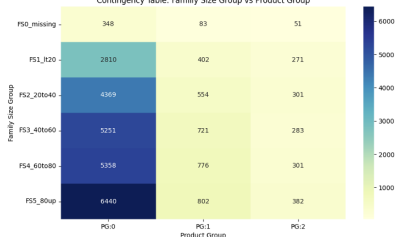
- **Patterns:**

- Economic Status Influence
- Variability in product premiums
- Correlations between customer age, family size, and product selection

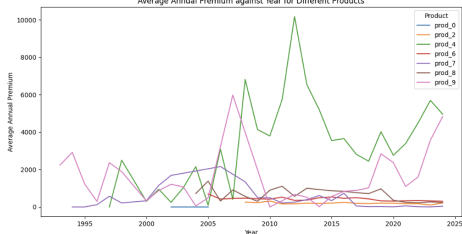
- **Challenges:**

- Fragmented Data
- High Dimensionality
- Advisor-Customer features Matching Complexity
- Lack of Match Quality Benchmark

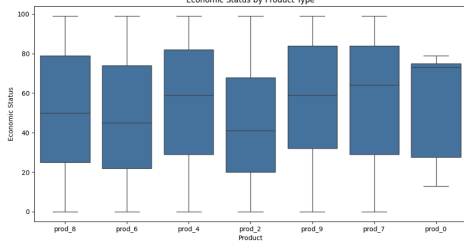
Contingency Table: Family Size Group vs Product Group



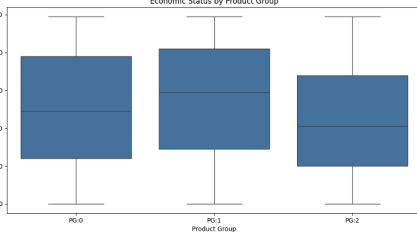
Average Annual Premium against Year for Different Products



Economic Status by Product Type



Economic Status by Product Group



Data Preprocessing

- **Merge Files:** Combined agent, client, and policy data.
- **Cleaning:** Removed duplicates and handled missing values.
- **Encoding:** One-hot encoding for non-ordered and ordinal encoding for ordered variables.
- **Feature Engineering:** Derived variables such as customer age groups and agent expertise.

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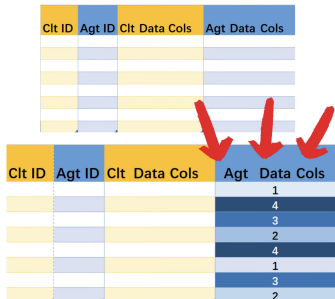
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Our Three-Phase Approach

- **Objective:** Match each client to the right financial advisor in three steps:
 - ① **Clustering:** Group advisors into “clusters” based on their attributes.
 - ② **Classification:** Assign each client to one of these clusters with a Random Forest.
 - ③ **Ranking:** Within that cluster, pick the best-performing agent using XGBoost.
- **Key Benefit:** This pipeline is scalable (handles many agents) and data-driven (uses past successes).

Phase 1: Clustering Agents

- **Goal:** Organize hundreds of agents into a few cohesive clusters.
- **Technique:**
 - Use Multiple Correspondence Analysis (MCA) to reduce high-dimensional data.
 - Apply a Gaussian Mixture Model (GMM) for flexible, probabilistic clustering.
- **Why It Helps:**
 - Simplifies the problem: handle “agent types” rather than every single agent.
 - Soft assignments let us account for agents who fit more than one profile.

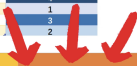


Clst ID	Agt ID	Clst Data Cols	Agt Data Cols

Clst ID	Agt ID	Clst Data Cols	Agt Data Cols
			1
			4
			3
			2
			4
			1
			3
			2

Phase 2: Classifying Clients to the Right Cluster

- **Challenge:** For each new client, we must decide which agent cluster is a good fit.
- **Solution: Random Forest Classifier**
 - Takes client features
 - Predicts the *cluster ID* that historically produced the highest conversion rate.
- **Why Random Forest?**
 - Handles categorical and numerical data with minimal tuning.
 - Offers strong performance and interpretability (feature importances).

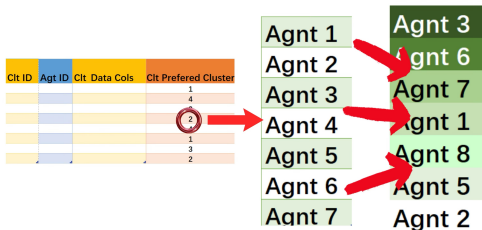


Clt ID	Agt ID	Clt Data Cols	Agt Data Cols
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			4
			3
			2
			4
			1
			3
			2

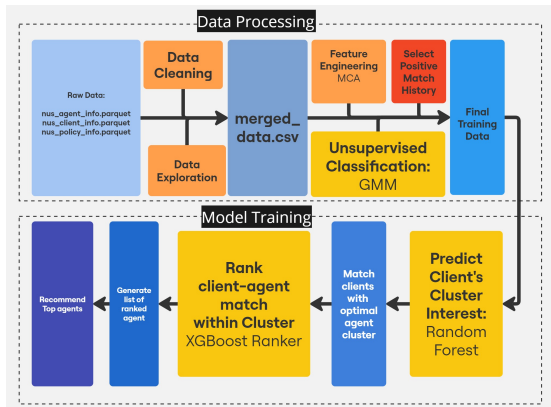
Clt ID	Agt ID	Clt Data Cols	Clt Preferred Cluster
			1
			4
			3
			2
			4
			1
			3
			2

Phase 3: Ranking Advisors Within Clusters

- **Within a cluster, multiple agents can match. We need the best one.**
- **XGBoost Ranking Model:**
 - Factors in agent *conversion success* and *in-force policies* to produce the score.
 - Optimize the ranking process for *learn* and *predict* the top-ranked agents.
- **Result:** Each client is routed to the highest-ranked (i.e., most suitable) agent within the predicted cluster.



Putting It All Together



End-to-End Flow:

- 1 **Cluster:** Group agents.
- 2 **Classify:** Map clients to clusters.
- 3 **Rank:** Pick the best agent in that cluster.

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Evaluation Metrics Overview

Metric	Definition / Meaning
Clustering	
Silhouette Score	Higher values indicate well-separated clusters.
Davies-Bouldin Index	Lower values indicate better clustering quality.
Log-Likelihood	Evaluates the goodness-of-fit for the clustering model.
Classification	
Accuracy	$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$
Precision	$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$
Recall	$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$
F1-score	$\text{F1_score} = 2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall})$
Ranking	
NDCG@5	Normalized Discounted Cumulative Gain at rank 5.
MRR	Mean Reciprocal Rank; average of the reciprocal ranks of the first relevant result for each query.

Results & Analysis

Metric	Value	Interpretation
Clustering		
Silhouette Score	Low (negative)	Overlapping clusters and poor separation.
Davies-Bouldin Index	High	Suggests that clusters are not well-separated (i.e., high intra-cluster variance).
Log-Likelihood	Moderate	Shows an adequate model fit, yet the clustering structure remains weak.
Classification		
Accuracy	~80%	Majority of client instances are correctly classified into clusters.
Precision & Recall	Balanced	Indicates consistent performance in identifying correct clusters.
F1-score	~79%	Demonstrates a good balance between precision and recall.
Ranking		
NDCG@5	1.0	Perfect ordering.
MRR	0.0033	First relevant advisor ranking is not good.

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Insights

- **Interpretable:**
 - Clear & Actionable – Ranker model insight: Optimized workload balance boosts efficiency & conversions
 - Fairness - Transparent model ensures bias detection & correction
 - Trust - Explainable recommendations increase policy purchases
- **Lightweight:** Minimal storage & easy deployment across branches

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Conclusion & Future Work

- **Conclusion:**

- Multi-stage model improves agent-client matching.
- Strong classification and ranking performance with room for clustering improvements.

- **Future Work:**

- Periodically retrain the model to ensure the data are up-to-date.
- Hyperparameter tuning and further feature engineering.
- Incorporate more industry specific knowledge to define more accurate metrics for “successful” transactions.

Thank You!

